Data Structures

Lecture 16
Tree (Basics)

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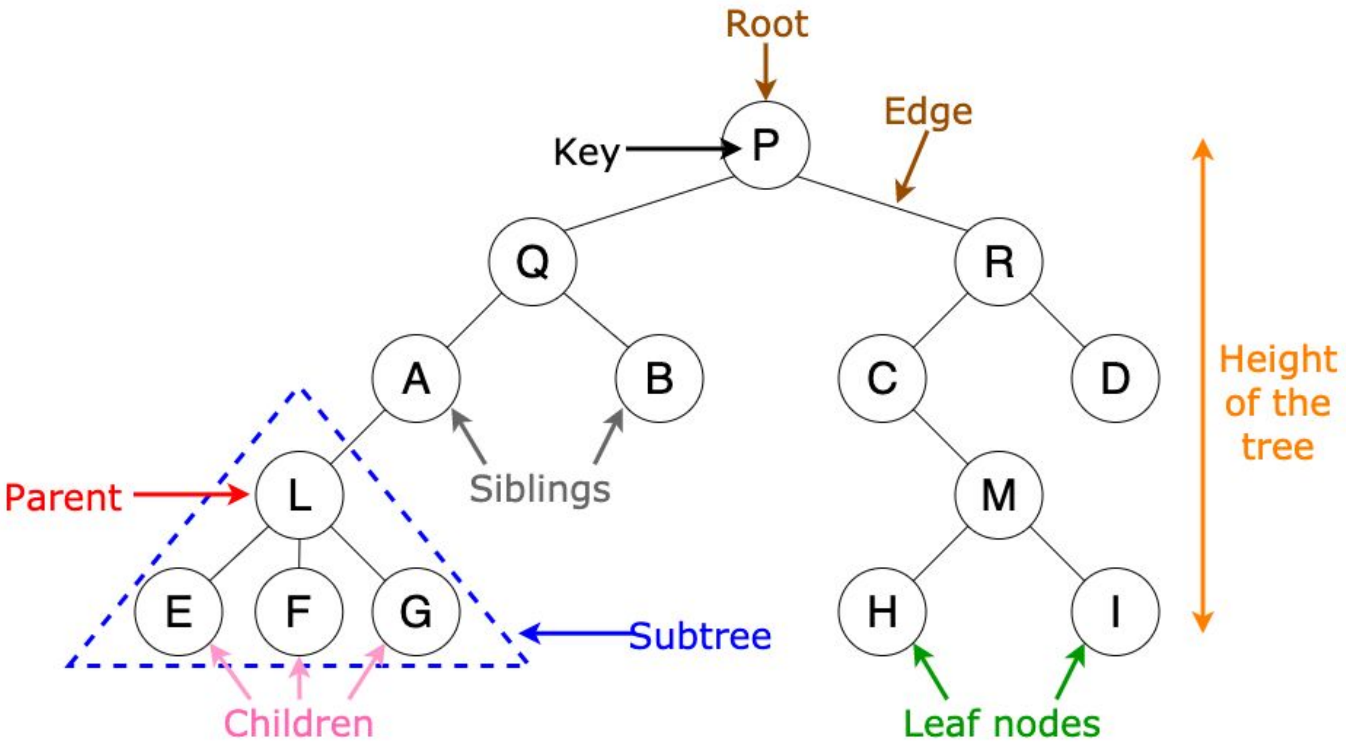
Trees



Root
Edge
Key
P
Q
R
A
B
C
D
L
M
E
F
G
H
I
Height of the tree
Parent
Siblings
Subtree
Children
Leaf nodes

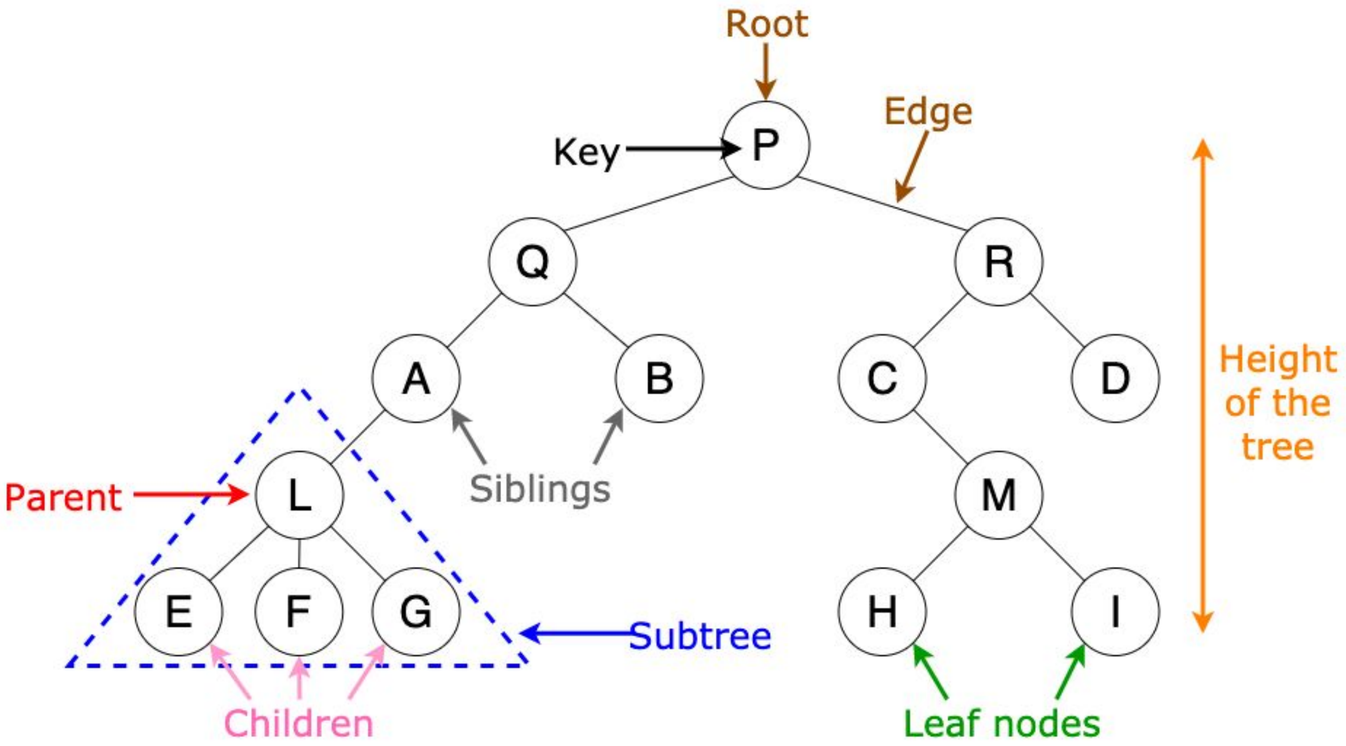
Why Trees?

Sorting New Elements

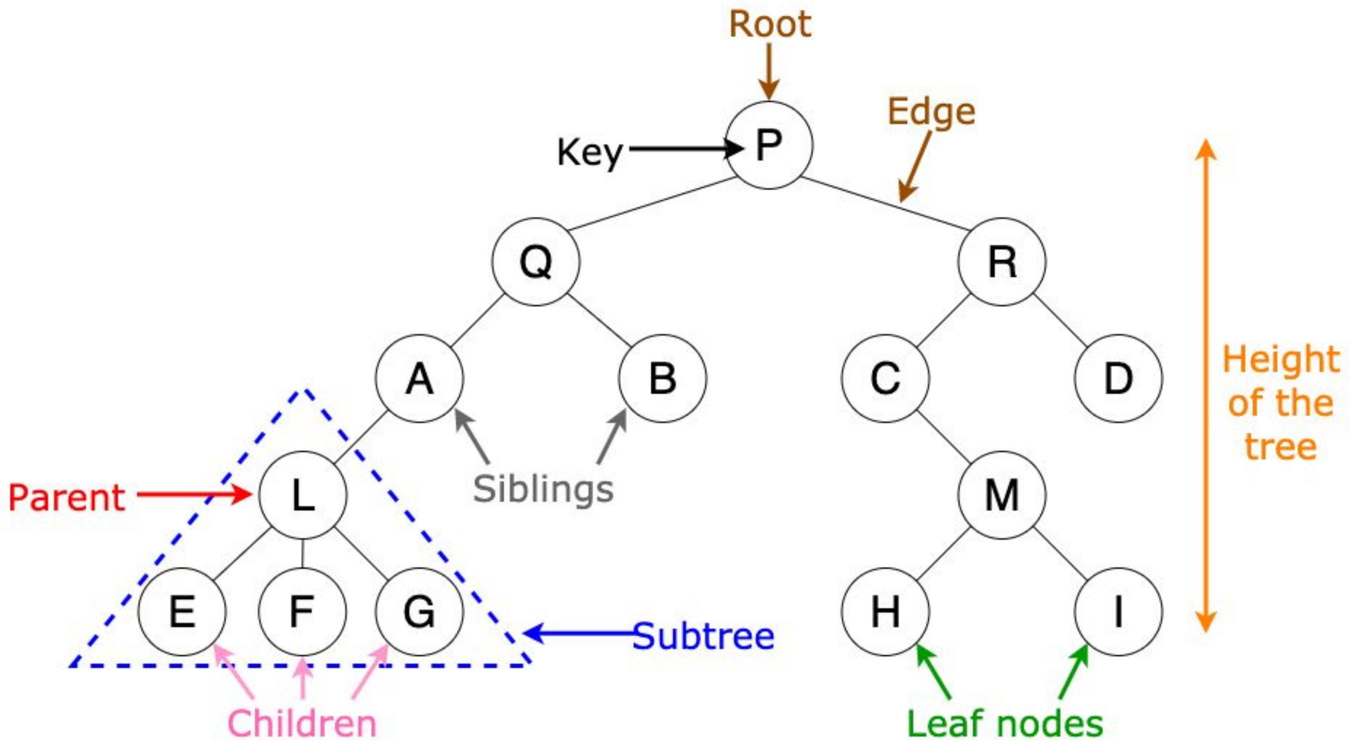
Folder/File System Structure

Computer Network Algorithms

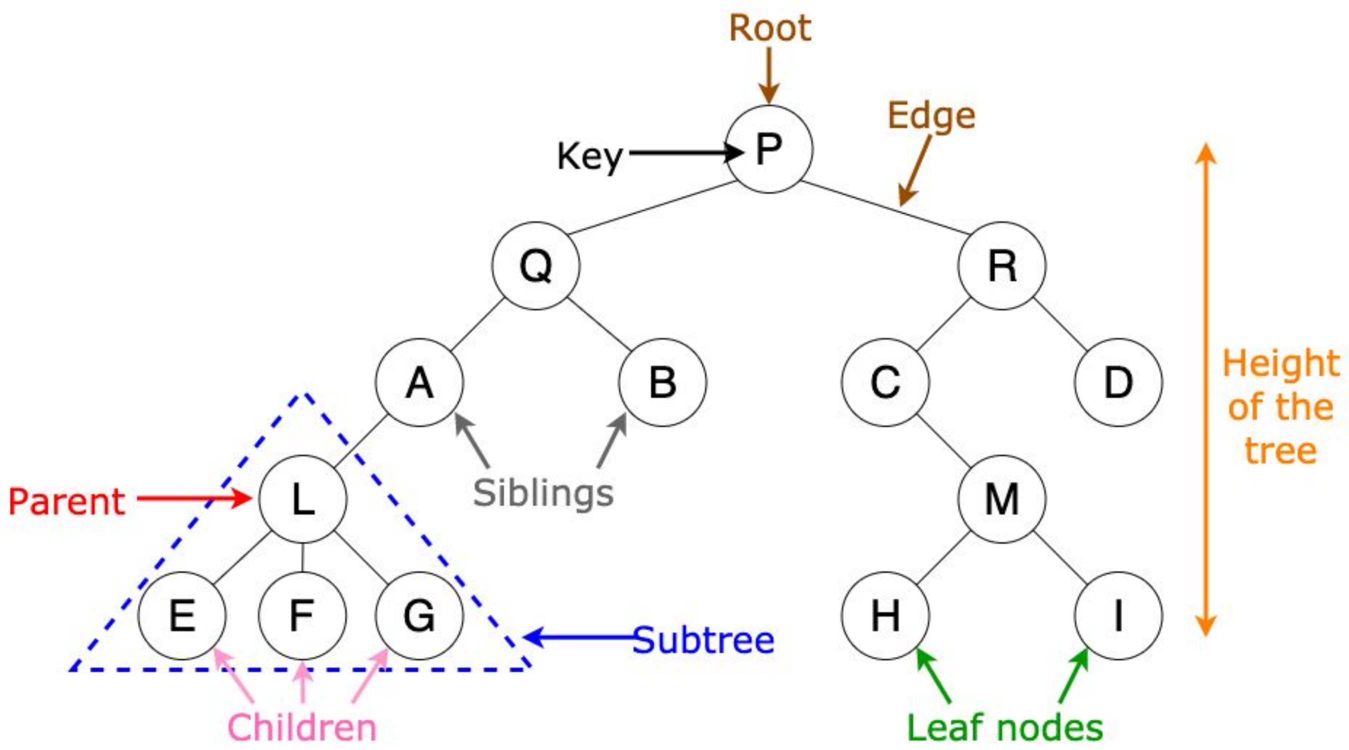
Trees - Root/Leaf/Non-Leaf



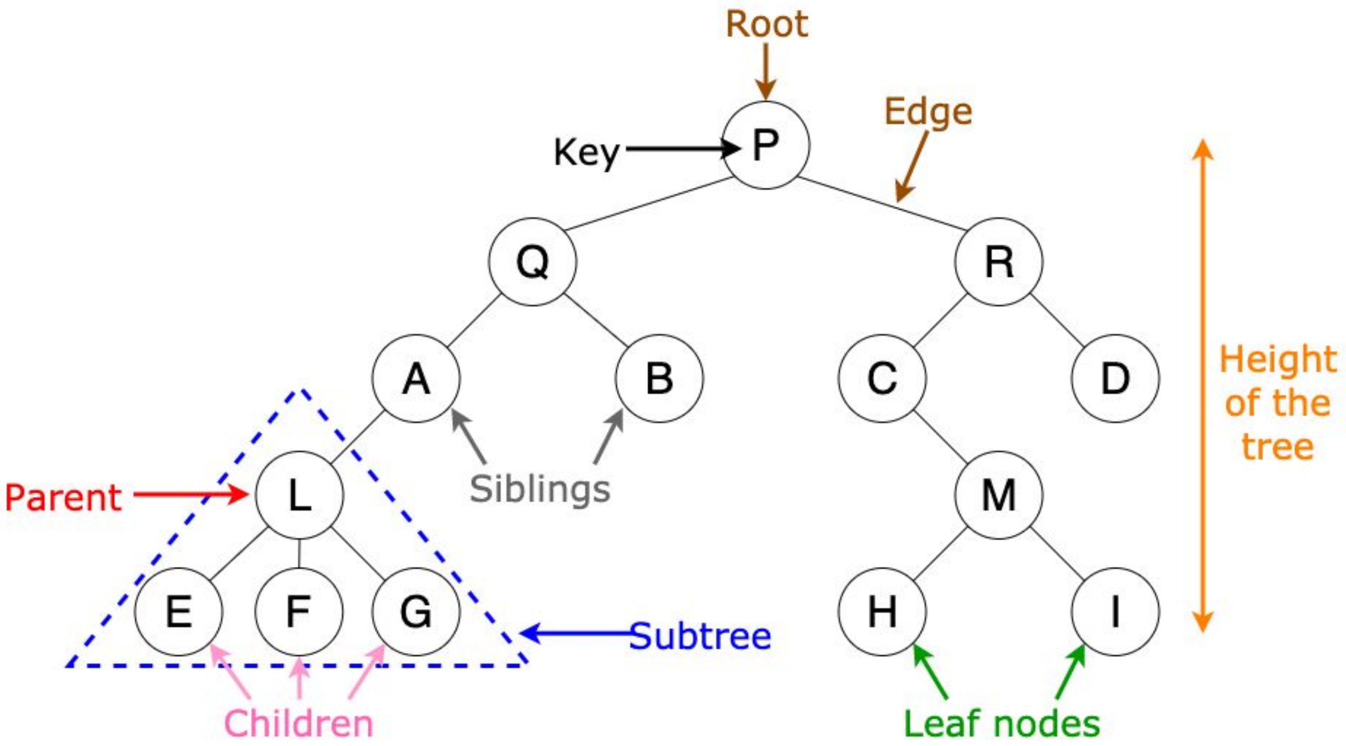
Trees - Parent/Child/Siblings



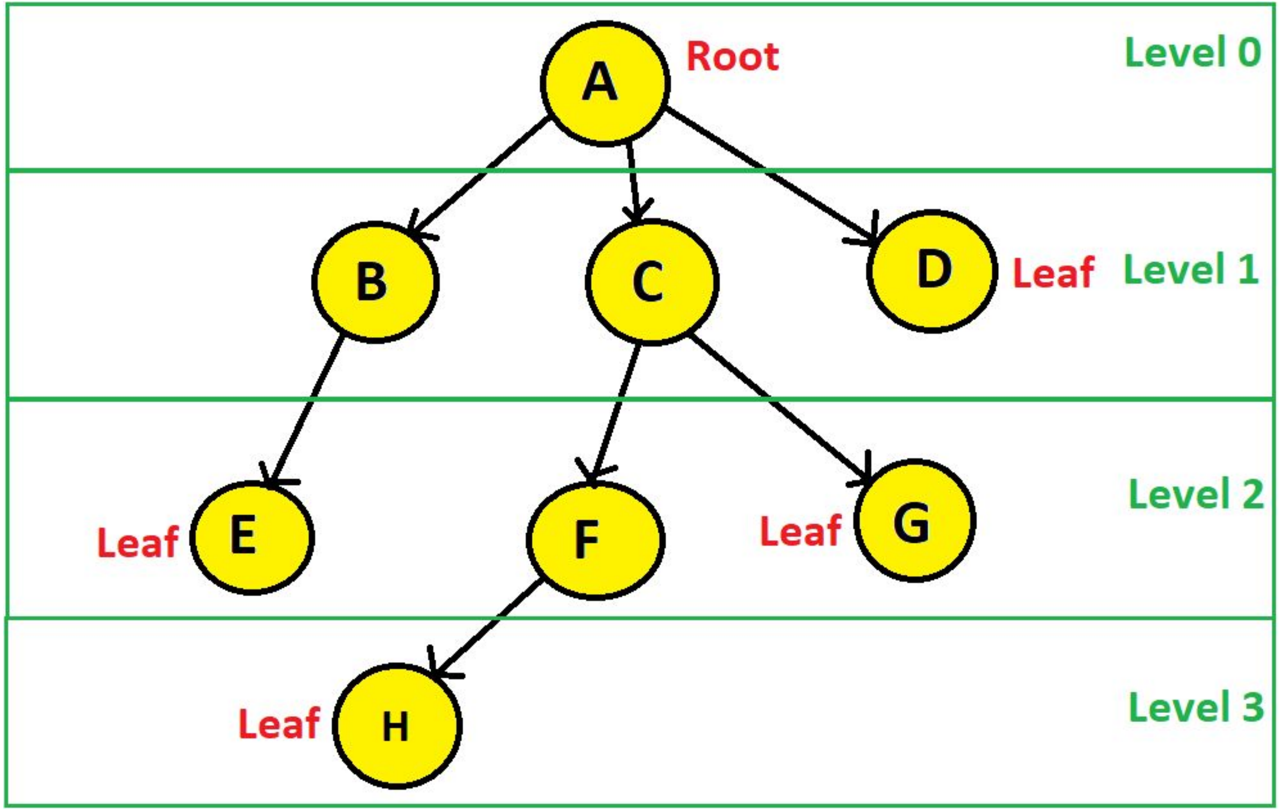
Trees - Edge/Path



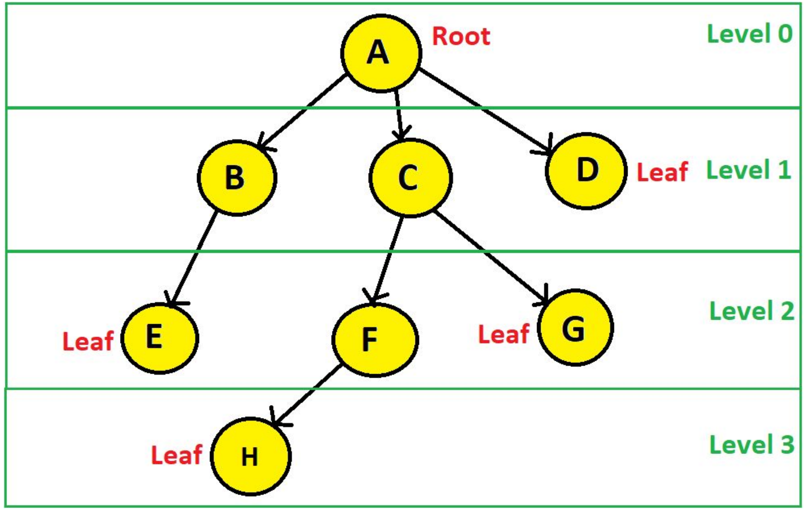
Trees - Degree



Trees - Depth/Height/Level



Trees - Depth/Height/Level



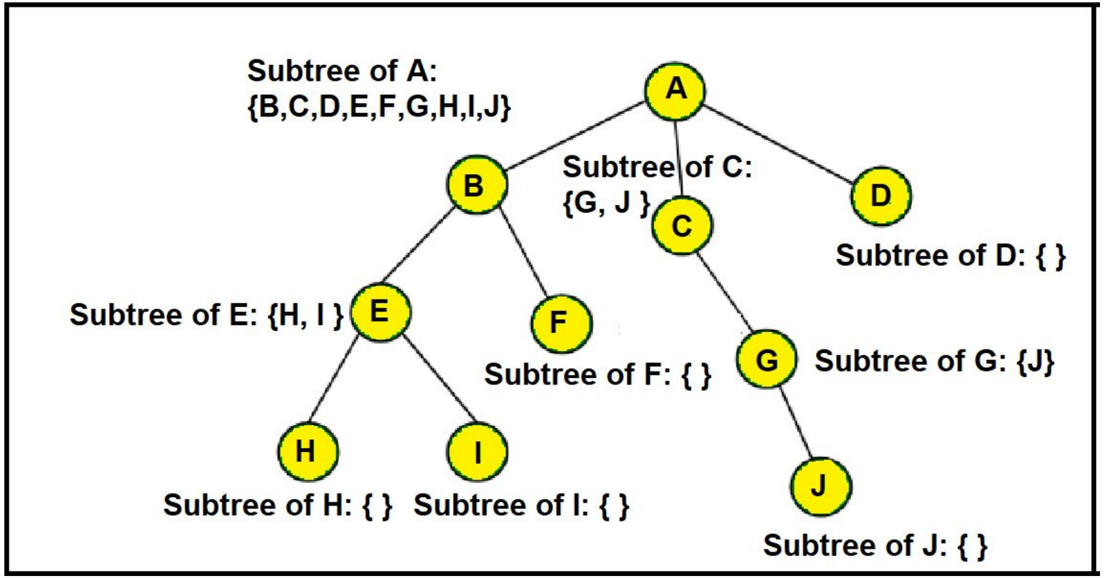
Depth of A?

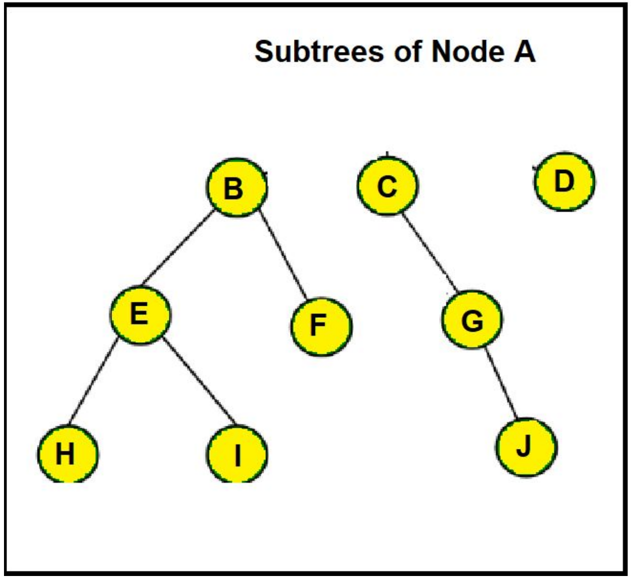
Height of A?

Depth != Height

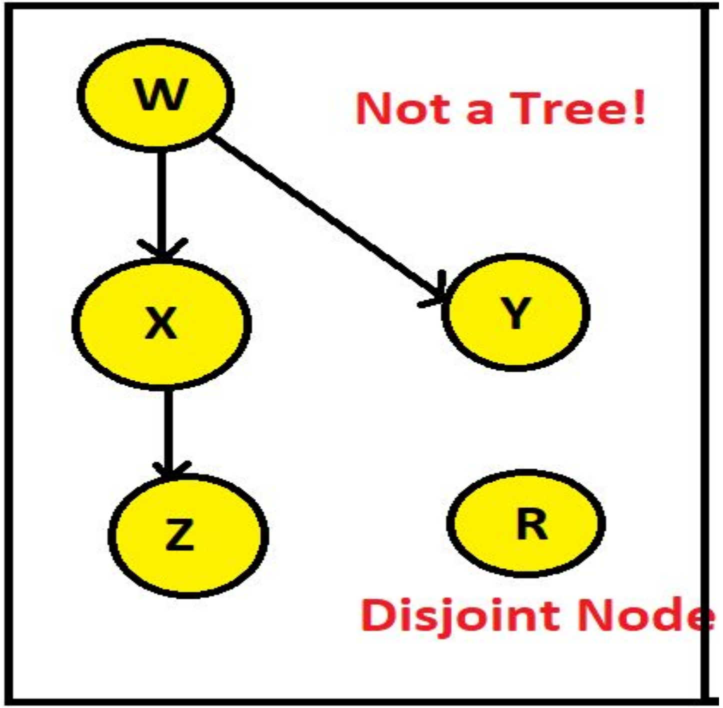
Depth = Level

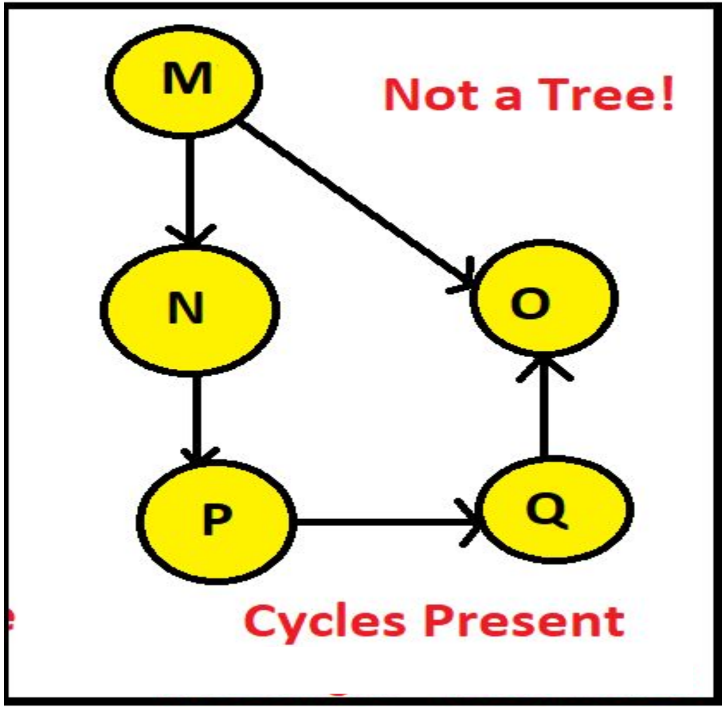
Trees - Subtree



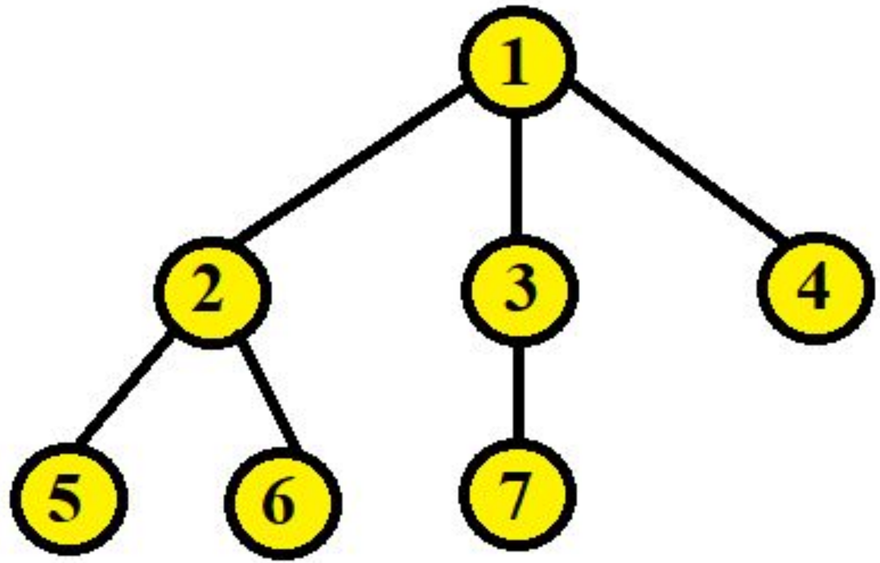


Trees - Characteristics





Trees - Build a Tree




```python
class Node:
    def __init__(self, elem):
        self.elem = elem
        self.children = []

root = Node(1)
root.children += [Node(2)]
root.children += [Node(3)]
root.children += [Node(4)]
root.children[0].children += [Node(5)]
root.children[0].children += [Node(6)]
root.children[1].children += [Node(7)]
```

Dynamic Representation with Linked List

Sequential Representation with Array